PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Estimation of sleep problems among pregnant women during COVID-19 pandemic: A systematic review and meta-analysis
AUTHORS	Alimoradi, Zainab; Abdi, Fatemeh; Gozal, David; Pakpour, Amir

VERSION 1 – REVIEW

REVIEWER	Miranda, Jezid Universidad de Cartagena, Obstetrics and Gynecology
REVIEW RETURNED	16-Aug-2021

GENERAL COMMENTS	The results of the manuscript are relevant. Previous studies assessing additional effects of the covid-19 pandemic have reported high levels of stress in pregnant women, which might also influence sleep quality and adverse consequences for the wellbeing of the pregnancy, including preeclampsia, preterm births, stillbirths, and low Apgar scores. The authors followed strict protocol for the study design that included criteria for study selection, using the PRISMA flowchart. In addition, the definition of sleep problems was based on validated instruments. The study found out that the pandemic might be associated with a significant increase in the rates of sleep disorders. Interestingly, the studies included registered the sleep disorder during the pandemic period. I believe that the manuscript is suitable for publication. Yet, there are some suggestions that I have for the authors: Any ideas regarding whether the risk of sleep disorders would be higher in pregnant women than in the general population? There is no data regarding which trimester might be more susceptible to develop sleep disorders. Previous studies have reported that sleep problems are significant in the third trimester compared to the first and the second. If the authors used data from papers that assess sleep quality BEFORE lockdown or the vaccination process started. In that case, it might not be helpful to support the conclusion of the manuscript. Can the authors expand in which moment of the pandemic the data was extracted for each country? The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88]. However, after imputation the reported prevalence was 13%. How was this imputation done? Why it affects so much the first estimation?

	Please consider the following recommendations in terms of grammar: Abstract: •Instead of using “valid and reliable measure,” consider replacing it by “a valid and reliable measure” •Instead of using “Due to probability of publication bias” consider replacing it with “Due to the probability of publication bias” •Consider rephrasing this sentence: All English, peer-reviewed papers published between December 2019 and July 2021 were included. Key points • Instead of using “during covid pandemic.” consider replacing it with “during the covid pandemic.” Introduction • Instead of using “birth weight” consider replacing it with “low birthweight.”

REVIEWER	Han, Ru Chinese Academy of Sciences
REVIEW RETURNED	23-Aug-2021

GENERAL COMMENTS	This study focused on the sleep problems among pregnant women during COVID-19 pandemic with the method of systematic review and meta-analysis is very important topic for the health of pregnant women and fetus. The pregnant women were facing the stress during the pandemic of COVID-19, such as the risk of COVID-19 infection than the pregnant women before the pandemic. They might worry about the health of baby and themselves. The sleep problem might be a prominent phenomenon for pregnant women. In the introduction, the researchers presented the study question of the pregnant women’s sleeping problem. On the base of the previous studies, which shows the relationship between pregnancy and poor sleeping, the pandemic of COVID-19 was assumed to compounded the sleeping problem. And the researchers suggested “Despite the impact of the COVID-19 on sleep quality and quantity 18-20, little is known about how sleep among pregnant women has been affected by this current crisis.” There are several studies on the topic, with which the researchers can organize the review and meta-analysis. On the other hand, the researchers could emphasize the characteristics of pregnant women’s sleeping problems, and the reason why the COVID-19 would enhance the effect or influence the specific aspect of their sleeping. This meta-analysis study complied with the PRISMA guidelines mostly and well-prepared, while the differences of searching syntax in different databases are very confusing. In the discussion part, “Another study in China conducted by Zhou et al.(2020) found that the prevalence of insomnia symptoms in pregnancy during the COVID-19 pandemic was 2.6%. These inordinately high prevalence figures raise the possibility that the gestational period is uniquely susceptible to sleep disruption by all of the environmental and lifestyle changes imposed by the pandemic control measures, even in the absence of infection.” Do the inordinately high prevalence figures refer to 2.6%?

REVIEWER	Azene, Abebaw Bahir Dar University College of Medical and Health Sciences, Epidemiology and Biostatistics
REVIEW RETURNED	25-Sep-2021

GENERAL COMMENTS	The authors write a good manuscript. However, some parts of the manuscript needs improvement. In the abstract part, there are two figures which indicats the prevalence of sleep problem which are 56% and 13%, so which one is the right figure? I have doubt about your search strategy, you might not achive your objective using those search term. Articles may used "covid 19", "SARS-2", "Novel Corona virus 2", etc, and also may use depression instead of "sleep". You used random effect model but I2 is 99% which indicate that you mixed Appel and orange. Which random effect model you employed? I have not seen heterogenous assessment like subgroup analysis in the manuscript, right? The discussion should be in line with your final report that means you should compare 13% with other study instead of 56%.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Jezid Miranda, Universidad de Cartagena
Comments to the Author:
The results of the manuscript are relevant. Previous studies assessing additional effects of the covid-19 pandemic have reported high levels of stress in pregnant women, which might also influence sleep quality and adverse consequences for the wellbeing of the pregnancy, including preeclampsia, preterm births, stillbirths, and low Apgar scores. The authors followed strict protocol for the study design that included criteria for study selection, using the PRISMA flowchart. In addition, the definition of sleep problems was based on validated instruments.
The study found out that the pandemic might be associated with a significant increase in the rates of sleep disorders. Interestingly, the studies included registered the sleep disorder during the pandemic period. I believe that the manuscript is suitable for publication. Yet, there are some suggestions that I have for the authors:

Any ideas regarding whether the risk of sleep disorders would be higher in pregnant women than in the general population?
Reply: Evidence from current systematic review is added:
“In a current systematic review, the pooled prevalence of sleep problems among the general population during the Covid-19 pandemic was estimated at 18% (95% CI: 15-21%) 22.”

There is no data regarding which trimester might be more susceptible to develop sleep disorders. Previous studies have reported that sleep problems are significant in the third trimester compared to the first and the second.
Reply: Data related to the items requested were not reported in the retrieved studies. That's the reason why we had no data to analyze or report. This point has been included in the limitations of study
“Also, data regarding which trimester might be more susceptible to develop sleep disorders was not reported within included studies. Consequently, we could not compare prevalence of sleep problems based on pregnancy trimesters”

If the authors used data from papers that assess sleep quality BEFORE lockdown or the vaccination process started. In that case, it might not be helpful to support the conclusion of the manuscript. Can the authors expand in which moment of the pandemic the data was extracted for each country?
Reply: The studies were performed at a time when the vaccination process had not yet begun. All studies were during the Corona pandemic, but quarantine policies were limited in various countries. Data collection time has been reported in studies. The data of the retrieved studies were not specifically at the time of quarantine, but as mentioned, all data studies were collected during the pandemic.

The pooled estimated prevalence of sleep problems was 56% [95% CI: 23 to 88]. However, after imputation the reported prevalence was 13%. How was this imputation done? Why it affects so much the first estimation?
Reply: Imputation was conducted via meta trim. Low number of studies and their probable overestimation might be the potential source of this change. Some explanation was added in Methods section:

“Meta-trim was used to correct probable publication bias 31. Meta-trim with Fill and Trim method was used to correct probable publication bias 31. Failure to correct the results in the presence of publication bias can lead to incorrect conclusions about the research question. Therefore, if the presence of publication bias is identified, the systematic reviewer should use the available methods to correct the results and present a more real conclusion. Although there are some methods recommended to correct publication bias 32, Fill and Trim method as a more conservative method was chosen and implemented.”
Please consider the following recommendations in terms of grammar:

Abstract:
•Instead of using “valid and reliable measure,” consider replacing it by “a valid and reliable measure”
•Instead of using “Due to probability of publication bias” consider replacing it with “Due to the probability of publication bias”
•Consider rephrasing this sentence: All English, peer-reviewed papers published between December 2019 and July 2021 were included.
Key points: Instead of using “during covid pandemic.” consider replacing it with “during the covid pandemic.”
Introduction: Instead of using “birth weight” consider replacing it with “low birthweight.”
Reply: Corrections are made as suggested.

Reviewer: 2
Dr. Ru Han, Chinese Academy of Sciences
Comments to the Author:
This study focused on the sleep problems among pregnant women during COVID-19 pandemic with the method of systematic review and meta-analysis is very important topic for the health of pregnant women and fetus. The pregnant women were facing the stress during the pandemic of COVID-19, such as the risk of COVID-19 infection than the pregnant women before the pandemic. They might worry about the health of baby and themselves. The sleep problem might be a prominent phenomenon for pregnant women.
In the introduction, the researchers presented the study question of the pregnant women’s sleeping problem. On the base of the previous studies, which shows the relationship between pregnancy and poor sleeping, the pandemic of COVID-19 was assumed to compounded the sleeping problem. And

the researchers suggested “Despite the impact of the COVID-19 on sleep quality and quantity 18-20, little is known about how sleep among pregnant women has been affected by this current crisis.” There are several studies on the topic, with which the researchers can organize the review and meta-analysis. On the other hand, the researchers could emphasize the characteristics of pregnant women’s sleeping problems, and the reason why the COVID-19 would enhance the effect or influence the specific aspect of their sleeping.
Reply:
Thanks for your comments. the introduction section was revised according to your suggestions.

This meta-analysis study complied with the PRISMA guidelines mostly and well-prepared, while the differences of searching syntax in different databases are very confusing.
Reply: search syntax was adapted based on advance search properties of different database. small difference in search syntax is due to this adaptation. The point is explained n method section.
“Search syntax was customized based on the advanced search attributes of each database.”

In the discussion part, “Another study in China conducted by Zhou et al.(2020) found that the prevalence of insomnia symptoms in pregnancy during the COVID-19 pandemic was 2.6%. These inordinately high prevalence figures raise the possibility that the gestational period is uniquely susceptible to sleep disruption by all of the environmental and lifestyle changes imposed by the pandemic control measures, even in the absence of infection.” Do the inordinately high prevalence figures refer to 2.6%?
Reply: Discussion is reorganized to better interpret the findings and compare it with existing evidence.
“We found that the overall prevalence of sleep problems during pregnancy in COVID-19 pandemic was nominally 56%, and after bias estimate corrections, 13%. In a recent systematic review, the pooled prevalence of sleep problems in the general population during the Covid-19 pandemic was estimated at 18% (95% CI: 15-21%) 22. Zhou et al.(2020) found that the prevalence of insomnia symptoms in pregnancy during the COVID-19 pandemic was 2.6% compared to 5.4% among non-pregnant women 8. Conversely, Xie et al.(2020) reported that 74.5% of women (not necessarily pregnant) complained about sleep problems in the COVID-19 pandemic 34. In another systematic review and meta-analysis, the corrected pooled estimated prevalence of sleep problems was 24% (95% CI: 19 to 29%) for female participants 23. Thus, sleep problems are prevalent among pregnant women during the COVID-19 pandemic, but not more frequent than in the general population. Some factors may account for such observations, namely a decision to become pregnant is usually taken during a period of better mental health and more secure financial situation. Therefore, pregnan women may have improved and more stable mental health condition than non-pregnant women. Second, pregnant women receive the focus of family attention at all times, and such uniue support mechanisms may be especially implemented by family members during COVID-19 epidemic. Third, increased contact with medical workers for their prenatal care can provide support and decrease stress symptoms 35. We surmise that all these factors may mitigate the driving factors for sleep problems and lead to less insomnia in pregnant women.”

Reviewer: 3
Mr. Abebaw Azene, Bahir Dar University College of Medical and Health Sciences
Comments to the Author:
The authors write a good manuscript. However, some parts of the manuscript needs improvement.
In the abstract part, there are two figures which indicats the prevalence of sleep problem which are 56% and 13%, so which one is the right figure?
Reply: 56% is estimated pooled prevalence based on primary analysis. As the publication bias was found, so we tried to correct the results using Meta-trim with Fill and Trim method. The corrected prevalence is 13%. through the text, it is mentioned that 13% is corrected pooled prevalence. The explanation is added in method section:

“Meta-trim was used to correct probable publication bias 31. Meta-trim with Fill and Trim method was used to correct probable publication bias 31. Failure to correct the results in the presence of publication bias can lead to incorrect conclusions about the research question. Therefore, if the presence of publication bias is identified, the systematic reviewer should use the available methods to correct the results and present a more real conclusion. Although there are some methods recommended to correct publication bias 32, Fill and Trim method as a more conservative method was chosen and executed.”
I have doubt about your search strategy, you might not achieve your objective using those search term. Articles may used "covid 19", "SARS-2", "Novel Corona virus 2", etc, and also may use depression instead of "sleep".
Reply: we used a wide syntax with main component of Exposure and Outcome based on PECO style. Covid-19 and its synonyms were considered as exposure and sleep problems and its synonyms were used as outcome. This wide syntax lead to find related words through all text. Also having data regarding sleep problem using valid and reliable measure was considered as inclusion criteria. When studies assess the sleep, even reporting data in table, we could retrieve it with this syntax. So this search syntax help us to have a comprehensive search. As you can see the search retrieved 7,263 in the first stage. All of the retrieved papers were assessed based on eligibility criteria. Explanation are added:
“Search syntax was customized based on the advanced search attributes of each database. Key search components were selected based on PECO search strategy (i.e., Patient/Problem, Exposure, Comparison and Outcome) 25 to answer the research question. In the present study, key elements of exposure (Covid-19) and outcome (sleep problem) were selected.”

You used random effect model but I2 is 99% which indicate that you mixed Apple and orange.
Reply: the "apples and oranges" problem mostly refer to summarizing studies with varying methodologies (Johnson, B. T., Scott-Sheldon, L. A., Snyder, L. B., Noar, S. M., & Huedo-Medina, T. B. (2008). Contemporary approaches to meta-analysis in communication research). But in present study we tried to select papers with same study design, same outcome, using two known valid and reliable measure for assessing the main outcome.

Which random effect model you employed? I have not seen heterogenous assessment like subgroup analysis in the manuscript, right?
Reply: Random effect was set through all analysis as random option for stata syntax. the results of meta- regression was provided in table 2. Due to low number of studies, meta regression was used instead of subgroup analysis.

Table 2. Results of meta-regression regarding estimated pooled prevalence
univariable meta-regression
Tau2 Adj. R2 (%) I2 res. (%) p S.E. Coefficient Variable
0.05 32.23 99.72 0.11 0.08 0.15 Country
0.04 64.01 98.76 0.07 0.05 -0.15 Age
0.09 -17.10 99.82 0.74 0.06 0.02 NOS- score
0.09 -3.06 99.61 0.41 0.22 0.20 Measure of sleep
multivariable meta-regression
0.04 63.91 98.61 0.42 0.09 0.08 Country
0.17 0.06 -0.12 Age

The discussion should be in line with your final report that means you should compare 13% with other study instead of 56%.
Reply: Discussion section is reorganized based on the comment and addressed the requested issues raised by the Reviewer.

VERSION 2 – REVIEW

REVIEWER	Han, Ru Chinese Academy of Sciences
REVIEW RETURNED	08-Feb-2022

GENERAL COMMENTS	It's necessary to present the explanations for the difference of results between Zhou et al.(2020) and Xie et al.(2020) in conclusion.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Dr. Ru Han, Chinese Academy of Sciences

Comments to the Author: It's necessary to present the explanations for the difference of results between Zhou et al.(2020) and Xie et al.(2020) in conclusion.

Response: we thank the reviewer for raising this point and have now added a commentary to this effect.